Unit Vectors

A unit vector has a magnitude of 1 with no units.
The unit vector $\hat{i}$ points in the $+x$ direction
j points in the $+y$ direction and $\hat{k}$ points in the $+z$ direction
Any vector can be expressed in terms of its components as

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

— Scalar product

The scalar product (also called the "dot product") of two vectors is

$$\vec{A}\cdot\vec{B} = AB\cos\phi \qquad \vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$$

— Vector product

The vector product (cross product) of two vectors has magnitude

$|\vec{A}\times\vec{B}| = AB\sin\phi$ and the right hand rule gives its direction.

CHAPTER 2 - Motion Along a Straight Line

— Displacement, time and average velocity

A object moving along the x-axis has a coordinate x
The change in the object's coordinate is $\Delta x = x_2 - x_1$
The average x-velocity of the hence is $v_{ave\text{-}x} = \Delta x / \Delta t$
The instantaneous velocity is the velocity at a specific instant of time or specific point along the path and is given by $v_x = dx/dt$
The average speed is not the magnitude of the average velocity!
The average acceleration is $a_{ave\text{-}x} = \Delta v_x / \Delta t$
The instantaneous acceleration is $a_x = dv_x/dt$

— The equations of motion with constant acceleration

$$v_x = v_{0x} + a_x\cdot t \qquad x = x_0 + v_{0x}\cdot t + \frac{1}{2}\cdot a_x\cdot t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0) \qquad x - x_0 = \left[\frac{v_{0x} + v_x}{2}\right]\cdot t$$

- Instantaneous velocity

$$v_x = \frac{dx}{dt} \qquad v_y = \frac{dy}{dt} \qquad v_z = \frac{dz}{dt}$$

- Velocity and position by integration

$$v_x = v_{0x} + \int_0^1 a_x \cdot dt \quad \text{and} \quad x = x_0 + \int_0^1 v_x \cdot dt$$

CHAPTER 3

MOTION IN 2D OR 3D

- Position vector

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \qquad \vec{v}_{ave} = \frac{\Delta \vec{r}}{\Delta t} \qquad \vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \text{ } \} \text{ instantaneous velocity vecto}$$

$$v_y^2 = v_{0_y}^2 + 2a_y(y - y_0) \qquad x - x_0 = v_{0x} \cdot t + \frac{1}{2} a_x t^2$$

CHAPTER 4

NEWTON'S LAWS OF MOTION

– Newton's First Law

Simply stated – "An object at rest tends to stay at rest, an object in motion tends to stay in uniform motion."

More properly, "A body acted on by zero net force moves with constant velocity and zero acceleration"

Newton's first law is valid only in non accelerating inertial frames.

– Common Types of Forces

- The normal force
- Friction force
- Tension force
- Weight

– Relative velocity in one dimension

If point P is moving relative to reference frame A, we denote the velocity of P relative to frame A as $v_{P/A}$

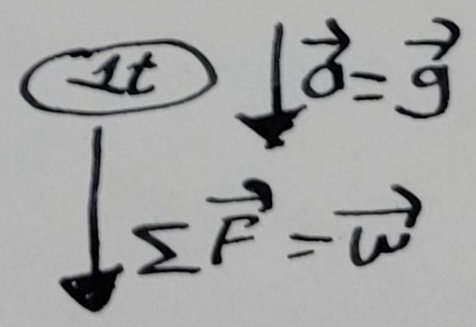
If P is moving relative to frame B and frame B is moving relative to frame A, then the x-velocity of P relative to frame A is

$$v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x}$$

– Mass and weight

The weight of an object (on the earth) is the gravitational force that the earth exerts on it.

$W = m \cdot g$ The value of g depends on altitude.



- Newton's Second Law

$$\sum \vec{F}_x = m \cdot a_x \qquad \sum \vec{F}_y = m \cdot a_y \qquad \sum \vec{F}_{Net} = m \cdot a$$

- Newton's Third Law

If you exert a force on a body, the body always exert a force (the "reaction") back upon you. So action = reaction

Reaction force has the same magnitude but direction is opposite.

These forces act on different bodies.

24/10/2019

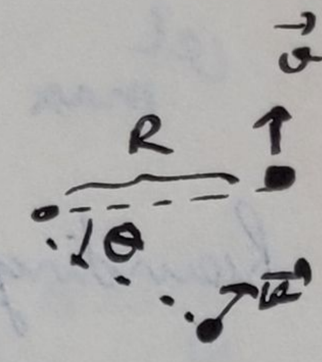
Uniform circular motion

ω : angular velocity ($\frac{\text{radian}}{\text{sec}}$) T : period (sec.) f : frequency (1/sec, Hz)

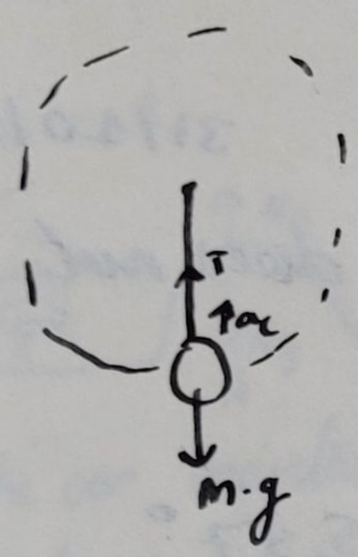
$\theta \rightarrow$ radian

1 radian $\cong 5,7^\circ$

$\omega = \frac{2\pi}{T} = 2\pi f$ $v = \omega . R$

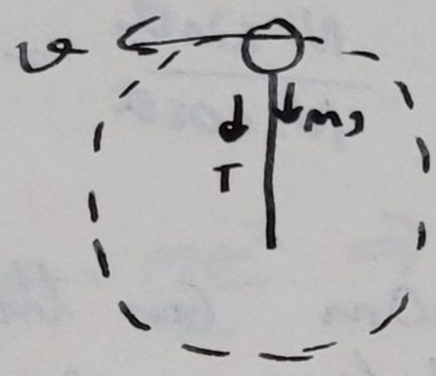


$F_c = m . a_c$ $a_c = \frac{v^2}{R} = \omega^2 . R$

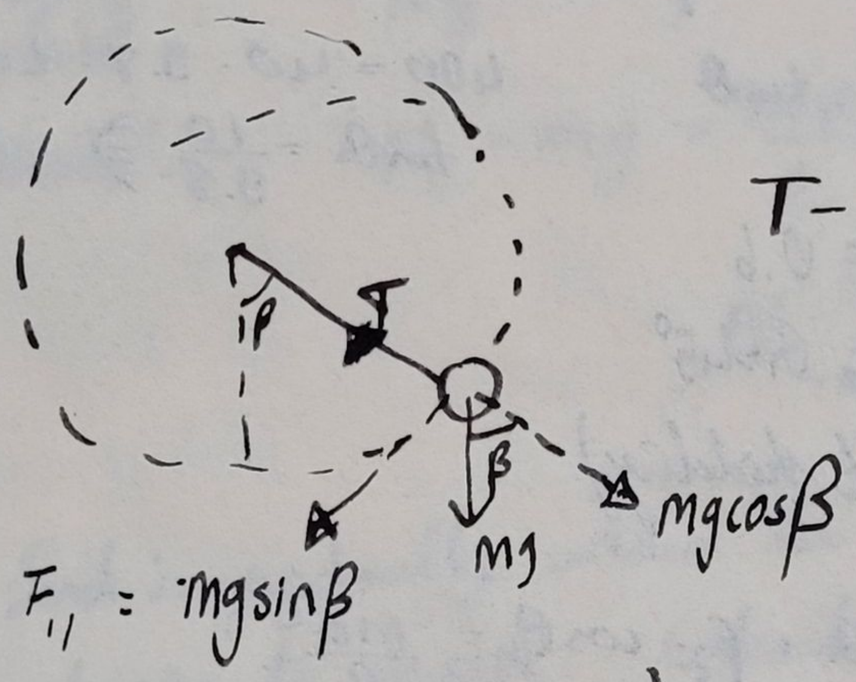


$T - m.g = m.a_c$

$T = m(g + a_c)$



$T + mg = F_c$



$T - mg\cos\beta = F_c$

$F_{//} = m . a_{//} = mg\sin\beta$

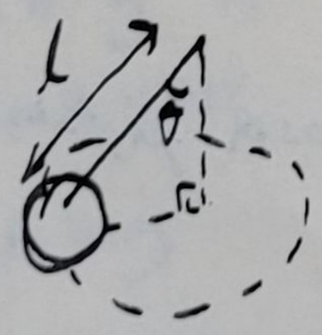
$a_{//} = g . \sin\beta$

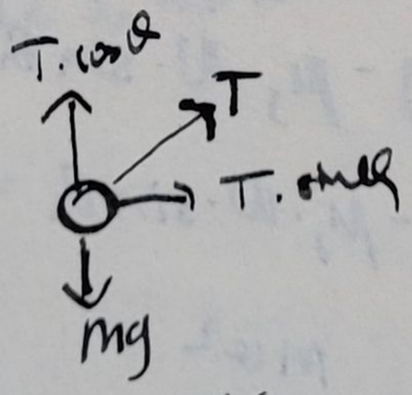
$a_{//} = \frac{dv}{dt}$

For non-uniform circular motion ($a_{//} \neq 0$)

$\vec{a} = \vec{a}_\perp + \vec{a}_{//}$ $|a| = \sqrt{(a_\perp)^2 + (a_{//})^2}$

Ex/ Find the acceleration of the ball? Find the period





$\Sigma F_{net} = ma_c$

$T . \sin\theta = \frac{mv^2}{l . \sin\theta}$

$T . \cos\theta = mg$

$a_c = \frac{v^2}{l . \sin\theta} = g . \tan\theta$

$\frac{T . \sin\theta}{T . \cos\theta} = \frac{v^2 . l}{g . \sin\theta . l}$

$v^2 = \frac{g l \sin^2\theta}{l \cos\theta}$

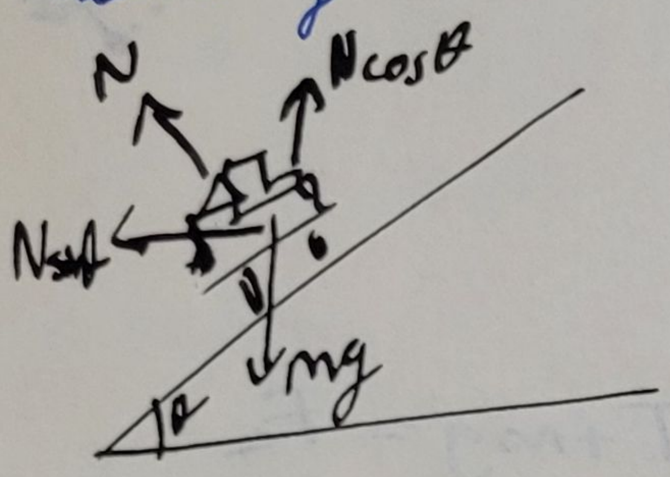
$T = \frac{2\pi l \sin\theta}{v}$

$T = 2\pi\sqrt{\frac{l}{g} . \cos\theta}$

Rounding an unbanked curve 1

$$v^2_{max} \leq \mu_s \cdot R \cdot g$$

Rounding a banked curve



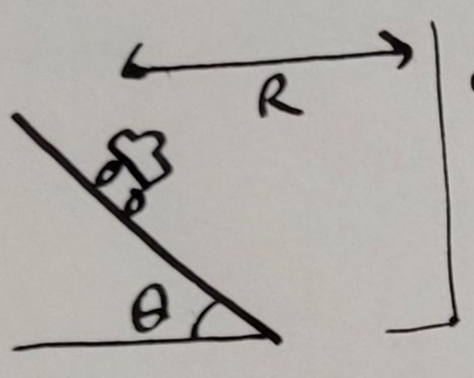
$y: N \cdot \cos\theta = mg \qquad f_s \neq 0$

$x: \Sigma F_{Net} = ma_c \qquad v_{max} = \sqrt{R \cdot g \cdot \tan\theta}$

$$N \cdot \sin\theta = \frac{mv^2}{R}$$

$$\frac{N \cdot \sin\theta}{N \cdot \cos\theta} = \frac{\frac{mv^2}{R}}{mg} \qquad v^2 = R \cdot g \cdot \tan\theta$$

31/10/19



a) $R = 40m$ Car travels with $v = 20 m/s$ does not skid (ignore friction). Find θ

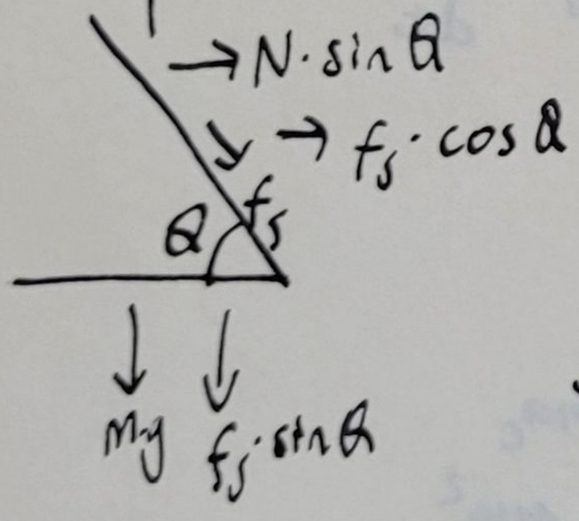
$v^2 = R \cdot g \cdot \tan\theta$

$400 = 40 \cdot 9.8 \cdot \tan\theta$

$\tan\theta = \frac{10}{9.8} \Rightarrow \theta \cong 45.57°$

b) Assume there is friction $\mu_s = 0.6$

What is the v_{max} of the car the $\theta = 45°$ make a safe turn? (without skidding)



$x: N \cdot \sin\theta + f_s \cdot \cos\theta = \frac{mv^2}{R}$

$$N \cdot \sin\theta + \mu_s \cdot N \cdot \cos\theta = \frac{mv^2}{R} \quad (1)$$

$y: N \cdot \cos\theta = mg + f_s \cdot \sin\theta$

$$N \cdot \cos\theta - m \cdot g - \mu_s \cdot N \cdot \sin\theta = 0$$

$$mg = N \cdot \cos\theta - \mu_s \cdot N \cdot \sin\theta = N(\cos\theta - \mu_s \cdot \sin\theta) \quad (2)$$

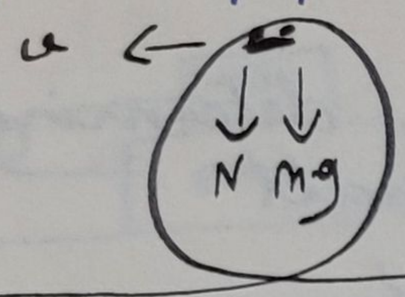
From (1)

$$N(\sin\theta + \mu_s \cdot \cos\theta) = \frac{mv^2}{R} \Rightarrow N = \frac{mv^2}{R \cdot (\sin\theta + \mu_s \cdot \cos\theta)}$$

From (2)

$$mg = \frac{mv^2}{R \cdot (\sin\theta + \mu_s \cdot \cos\theta)} \cdot (\cos\theta - \mu_s \sin\theta)$$

Loop the loop problem



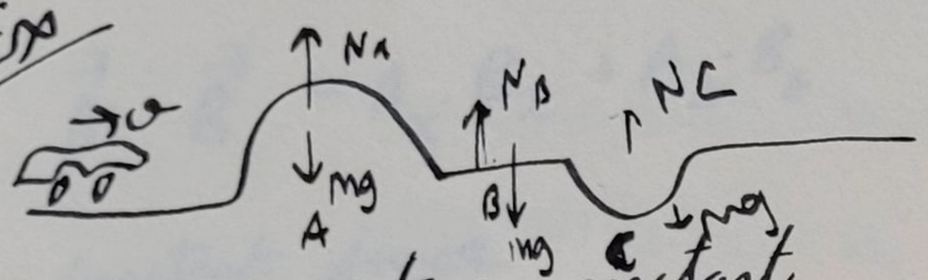
Calculate N?

$\sum F_{net} = m \cdot a_c \qquad N + m \cdot g = \frac{m v^2}{R} \qquad N = \frac{m v^2}{R} - mg \qquad N \geq 0$

If $N=0$ we can find v_{min}

$N = m\left(\frac{v^2}{R} - g\right) = 0 \Rightarrow \frac{v^2}{R} = g \Rightarrow v = \sqrt{g \cdot R}$

Ex



The car maintains a constant speed v during its travel.

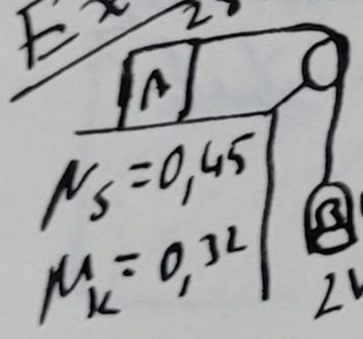
$mg - N_A = \frac{m v^2}{R} \Rightarrow N_A = mg - \frac{m v^2}{R}$

$N_B = m \cdot g$

$N_C - mg = \frac{m v^2}{R} \Rightarrow N_C = mg + \frac{m v^2}{R}$

$N_C > N_B > N_A$

Ex



Add sand to the bucket

Sand is gradually added to the bucket until system begins to move. Calculate the mass of the sand added to the bucket?

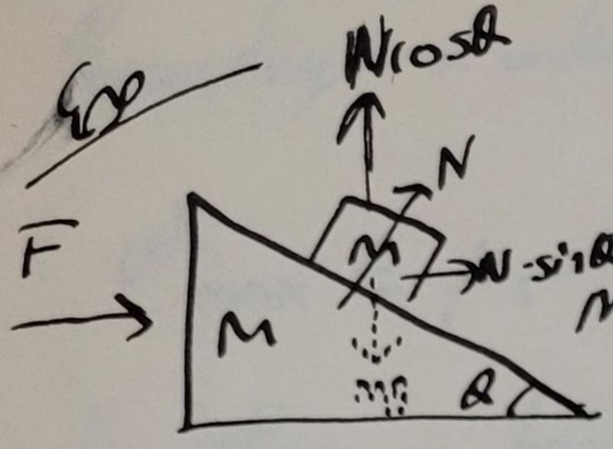
$f_{s_A} = 28 \cdot g \cdot 0,45 = 123,48\,N$

$m \cdot g = 123,48\,N$

$\hookrightarrow 12,6\,kg$

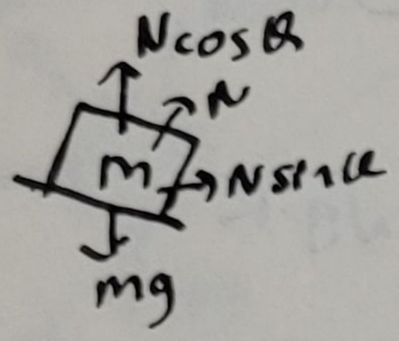
$12,6 - 2,0 = 10,6\,kg$

Ex

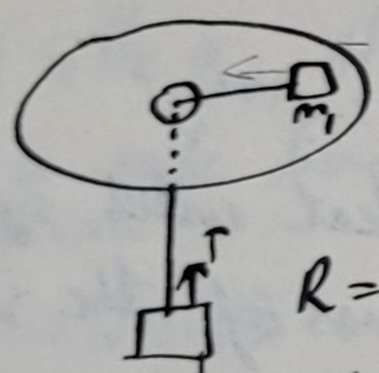


m and M are at rest

If μ_s is static friction coefficient, determine minimum horizontal force F so small block start to move up incline.



Ex



Disk rotates with $\omega = 4 rad/sec$ $m_1 = 1kg$ is located at a distance $R = 50cm$ away from center. $g = 10 m/s^2$ if $\mu_s = 0,4$

a) What is the minimum m_2 so that m_1 does not start sliding outward?

b) What is the maximum m_2 so that m_1 does not start sliding inward?

a) $T + F_f = m_1 \omega^2 R = 8N$

$T + \mu_s \cdot 0,4 \cdot 10 = 8N$

$T = 4N$

$T = m_2 g = 4N$

$m_2 = 0,4kg$

b) $T - F_f = m\omega^2 R = 8N$

$F_f = 4N$

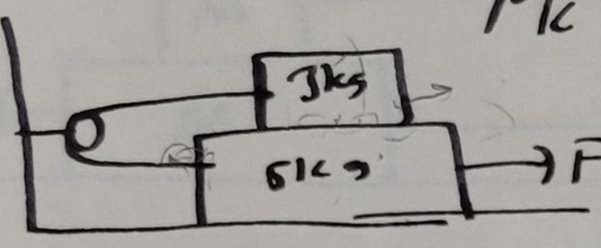
$T = 12N$

$12N = m_2 \cdot g$

$m_2 = 1,2kg$

Ex

84



$\mu_s = 0,6$

$\mu_k = 0,4$

What is the F_{min} to move two blocks?

Work and Energy

05/11/2019

- Vector multiplication

Scalar product (dot product)

$\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos\alpha$ $\quad \hat{i} \cdot \hat{i} = |\hat{i}| |\hat{i}| \cdot \cos 0 = 1$

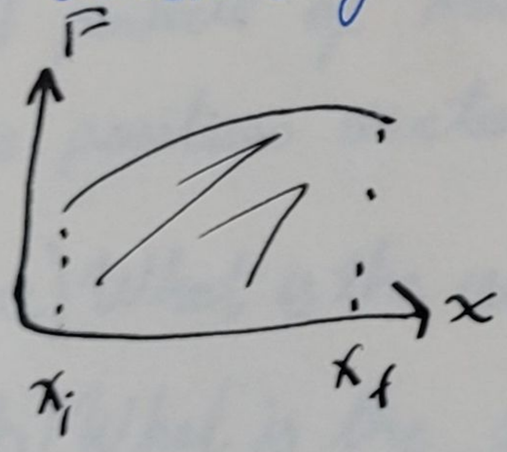
$\vec{A} \cdot \vec{B} = A_x \cdot B_x + A_z \cdot B_z$

- Constant force

$K.E = \frac{1}{2} m v^2$ $\quad F \cdot \Delta x = \Delta KE = \frac{1}{2} m (v_f^2 - v_i^2)$

$Work = \vec{F} \cdot \Delta \vec{x} = |\vec{F}| \cdot |\vec{x}| \cdot \cos\theta$

- Variable force



$$W = \int_{x_i}^{x_f} \vec{F}(x) \cdot d\vec{x}$$

07/11/2019

A car starts from rest and moves with constant $a = 2 m/s^2$ until it reaches $v = 26 m/s$ it then continues to move with constant speed.

a) For how long does the car accelerate?

$26 m/s = t \cdot 2$

$t = 13 s$

b) How much distance is covered by car during it accelerates?

$x = \frac{1}{2} \cdot 2 \cdot 13^2 = 169 m$

c) After reaching $v = 26 m/s$ how much longer should the car travel until its average speed comes $\bar{v} = 24 m/s$.

$x_0 = 169 m \quad t_1 = 13 s$

$x = 26 \cdot t_2$ $\quad \frac{169 + 26 \cdot t_2}{13 + t_2} = \frac{24 m}{1 s}$

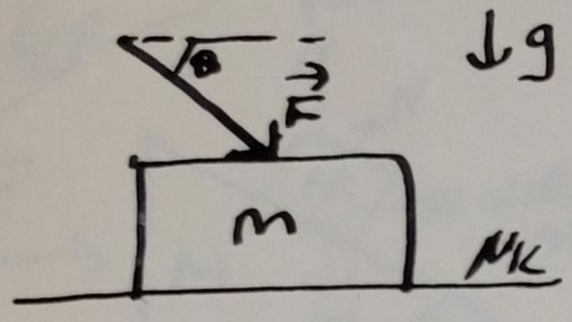
$169 + 26 t_2 = 24 t_2 + 24 \cdot 13$

$2t_2 = 13 \cdot 24 - 13 \cdot 13$

$2t_2 = 13(24 - 13) = 13 \cdot 9$

$t_2 = 58.5 s$

$$\mu_k = \mu_s = \mu$$



a) Find the expression for the magnitude of force needed to push the box at constant speed.

$a = 0 \Rightarrow \sum F_{net} = 0$

$F \cdot \cos\theta = N \cdot \mu$

~~$N \cdot \cos\theta = mg \cdot \mu$~~

$x: \sum F_x = m \cdot a_x$

$F \cdot \cos\theta - f_f = m \cdot 0$

$y: \sum F_y = 0$

$N - mg - F\sin\theta = 0$

~~$F\sin\theta = mg - N$~~

$F = \dfrac{mg - N}{\sin\theta}$

$N = mg + F\sin\theta$

b) Find the critical angle θ above which the becomes impossible to push?

$F \cdot \cos\theta = 0$

$\cos\theta = 0 \Rightarrow \theta = 90°$

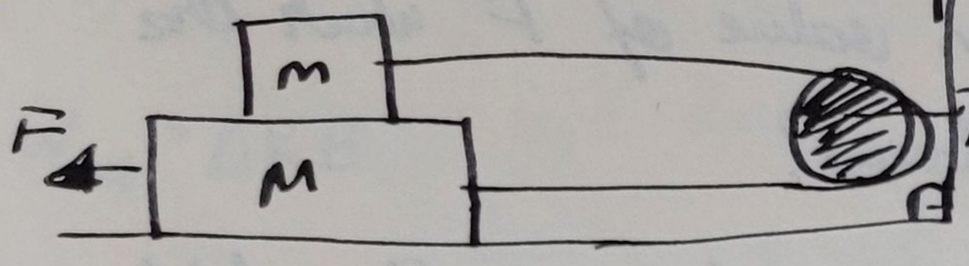
$F = \dfrac{\mu mg}{\cos\theta_c - \mu\sin\theta_c}$

$\cos\theta - \mu\sin\theta_c = 0$

$\mu = \cot\theta_c$

$\theta_c = \tan^{-1}(\mu)$

mass m rests initially. There is no significant friction.



a) What is the acceleration of top block with respect to the horizontal surface?

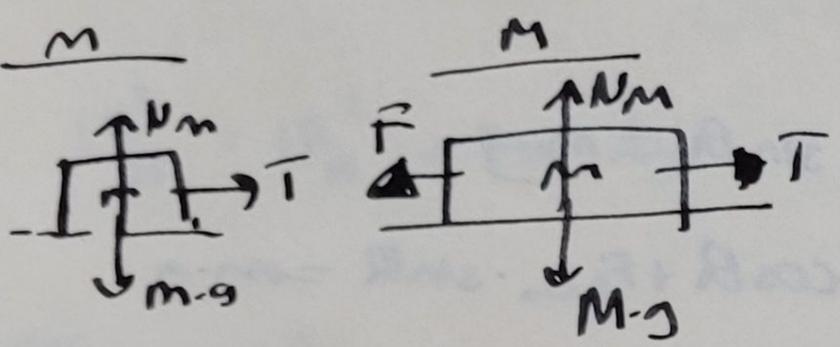
b) What is the acceleration of the top block relative to bottom block?

c) Tension is the cord?

$\vec{a}_{top} = \vec{a} - (-\vec{a}) = 2\vec{a}$

$T = m \cdot a$

$T = m\left(\frac{F}{m+M}\right)$



$T = m \cdot a$

$N_m = m \cdot g$

$F - T = M \cdot a$

~~$T = M \cdot a = T$~~

$F - ma = M \cdot a$

$F = a(m+M)$

$a = \frac{F}{m+M}$

$8\sin(5t)\cdot\cos 5t\cdot\hat{j} + 2\cos 10t \cdot 10$

- A particle of mass $m = 1kg$ is performing uniform circular motion. Its position vector is $\vec{r}(t) = 4\sin^2(5t)\hat{i} + \overbrace{4\sin(5t)\cos(5t)}^{2\sin(10t)}\hat{j}$

a) What is the velocity of the particle as a function of time?

b) What is the speed of particle?

c) What is the acceleration of particle?

d) What is the magnitude of net force?

e) What is the period of the motion?

f) What is the radius of circle?

a) $\vec{v} = 40\sin(5t)\cdot\cos 5t + 20\cos 10t = 20(\sin(10t)\hat{i} + \cos(10t)\hat{j})$

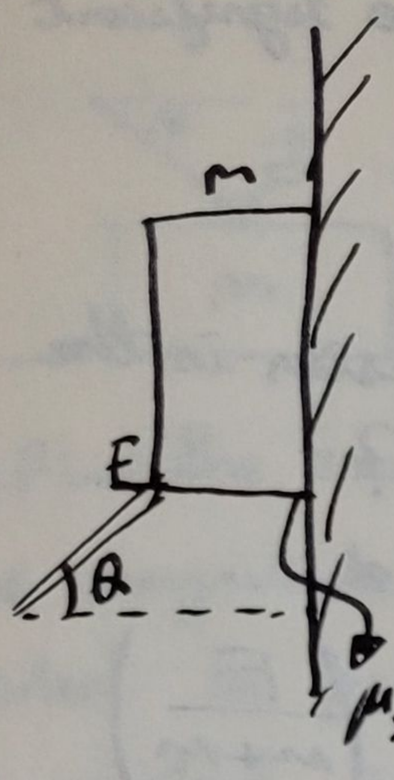
b) $|v| = 20 m/s$

c) $200\cdot(\cos(10t)\hat{i} - \sin(10t)\hat{j})$ $|a| = 200 m/s^2$

d) $|\vec{F}| = m\cdot\vec{a} = 1\cdot 200 = 200N$

e) ~~$mw^2r = m\cdot a$~~ $w = \frac{2\pi}{r}$ $\left(\frac{2\pi}{T}\right)^2\cdot 2 = 200$ $\frac{2\pi}{T} = 10\pi$ $T = \frac{\pi}{5}$

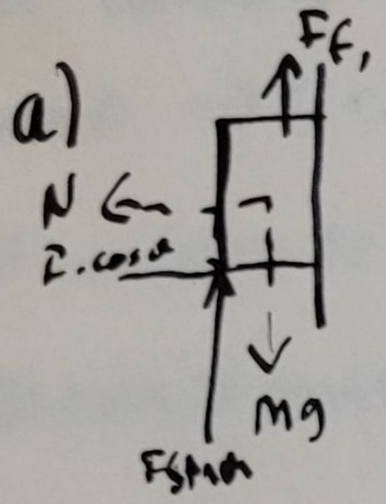
f) $r = \sqrt{x^2+y^2}$ $\frac{v^2}{r} = a$ $\frac{400}{r} = 200$ $r = 2m$



a) What is the minimum value of $\vec{F}$ which the block does not slide down?

b) What is the maximum value of F which the block does not slide?

a)



$$F_{f_1} + F_{max}\sin\theta = m\cdot g \rightarrow \mu_s \cdot N + F_{max}\cdot \sin\theta = m\cdot g$$

$$N - F_{max}\cos\theta = 0$$

$$N = F_{max}\cdot\cos\theta$$

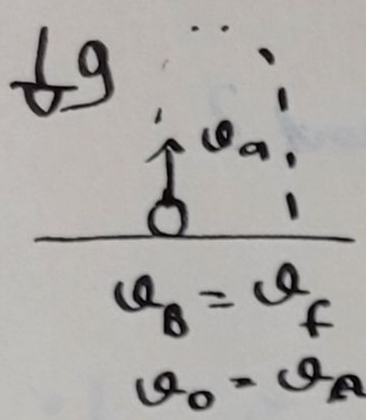
$$\mu_s \cdot F_{max}\cdot\cos\theta + F_{max}\cdot\sin\theta = m\cdot g$$

$$F_{max} = \frac{mg}{\mu_s\cdot\cos\theta + \sin\theta}$$

b)

19/11/2019

$W_{net} = \Delta KE$



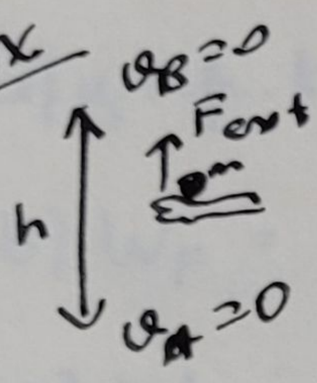
$$W_{net} = \Delta KE = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_0^2$$

$$W_{net} = -\frac{1}{2} m v_0^2 = \vec{F_g} \cdot \Delta y$$

$$|mg| \cdot |h| \cdot \cos 180° = -\frac{1}{2} \cdot m \cdot v_0^2 \qquad v_0^2 = 2 \cdot g \cdot h$$

$$v_B^2 = v_A^2 + 2a \cdot \Delta y$$

ex



$$W_{ext} + W_{gravitational} = 0$$

$$\int \vec{F}_{ext} \cdot \Delta y + \vec{F_g} \cdot \Delta y = 0$$

$$|mg| \cdot |h| \cdot \cos 0 + |mg| |h| \cos 180 = 0$$

$$mgh - mgh = 0$$

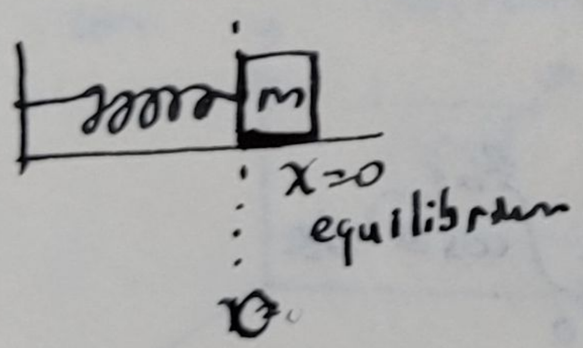
Work in 3D

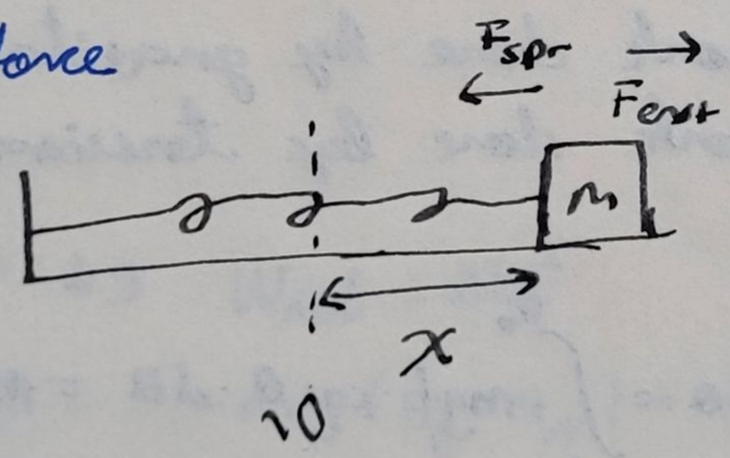
$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\vec{F} = F_x \cdot \hat{i} + F_y \cdot \hat{j} + F_z \cdot \hat{k}$$

$$W_{net} = \int_A^B \vec{F} \cdot d\vec{r} = \int_{x_A}^{x_B} F_x \cdot dx + \int_{y_A}^{y_B} \vec{F_y} \cdot \vec{d_y} + \int_{z_A}^{z_B} \vec{F_z} \cdot \vec{d_z}$$

- work done by spring force



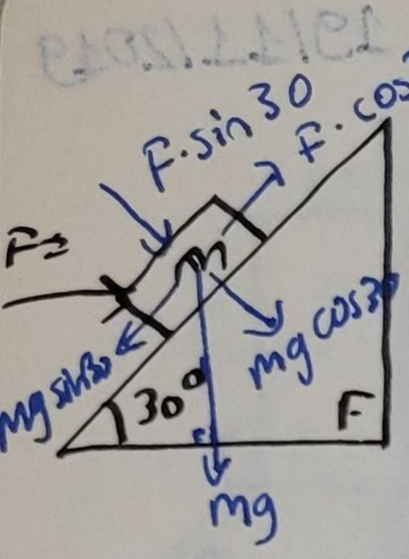


$$F_S = -k \cdot x = -F_{ext}$$

$$W_S = \int_0^x \vec{F_S} \cdot d\vec{x}$$

$$\boxed{W_S = -\frac{1}{2} \cdot k \cdot x^2}$$

$$W = -\int_0^x k \cdot x \, dx \quad \text{(stretching or compressing)}$$



$F = 150N$

$m = 18kg$

a) How much work is done by F when force pushes the block 5m up?

b) How much work is done by gravitation?

c) What is speed of block after displacement? (initially at rest)

a) $W_F = F \cdot \cos 30 \cdot 5 + F \cdot \sin 30 \cdot 5 \cdot \cos 90 = 650j$

b) $W_G = mg \sin 30 \cdot d \cdot \cos 180° + mg \cos 30 \cdot \cos 90$

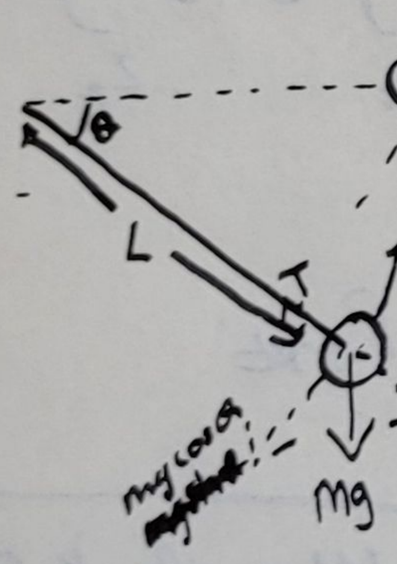
$= 18 \cdot 10 \cdot \frac{1}{2} \cdot 5 \cdot (-1) = -450j$

c) $W_{net} = \Delta KE = \frac{1}{2} m v_f^2 - \cancel{\frac{1}{2} m v_0^2}^{0} = W_F + W_G + W_N$

$\frac{1}{2} \cdot 18 \cdot v_f^2 = 650 - 450 + \cancel{W_N}^{0} = 200j$

$v_f = \sqrt{\frac{400}{18}} \approx 4 \, m/s$

21/11/2019



Mass released from horizontal late brought to $\theta = \theta_0$

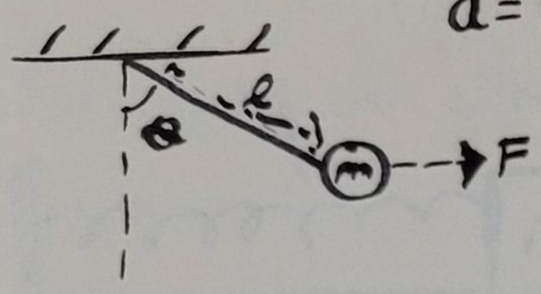
a) Find the work done by gravitation in terms of θ_0.

b) Find the work done by tension.

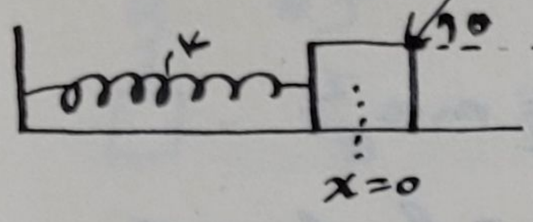
$$W_G = \int \vec{F}_G \cdot d\hat{\ell} = \int |mg| \cdot |L \cdot d\theta| \cdot \cos\theta = \int_0^{\theta_0} mg \cdot L \cdot \cos\theta \cdot d\theta = mgL \int_0^{\theta_0} \cos\theta \cdot d\theta$$

$$= mgL \sin\theta_0$$

$$W_T = \int_0^{\theta_0} \vec{T} \cdot d\vec{\ell} = \int |T| \cdot |d\ell| \cdot \cos 90 = 0$$



$a = 0 \quad W_G = ?$



F pushes the mass for a distance x until it comes to rest.

a) How much PE is in the spring? b) W_F? c) W_f?

d) Find x in terms of F, k, θ, m

$N = mg + F\sin\theta$

a) $U_s = \frac{1}{2} \cdot k \cdot x^2$

b) $W_F = F\cos\theta \cdot x \cdot \cos 0 = F \cdot \cos\theta \cdot x$

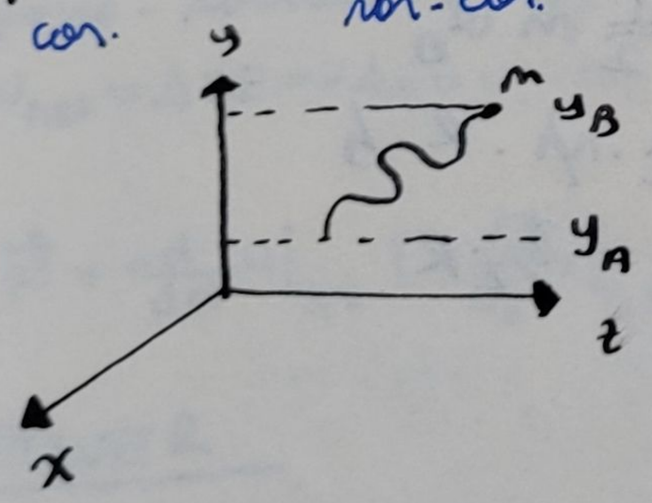
c) $W_{F_f} = \vec{F_f} \cdot \vec{x} = F_f \cdot x \cdot \cos 180 = -\mu_k \cdot (mg + F\sin\theta) \cdot x$

d) $W_{net} = \Delta KE \qquad W_G + W_{F_f} + W_F + W_S = \Delta KE = 0$

$$0 - \mu_k(mg + F\sin\theta) + F \cdot \cos\theta \cdot x - \frac{1}{2} \cdot k \cdot x^2 = 0$$

Conservative Forces

F(x) con. F(θ) non-con.



$W_{net} = \Delta KE$

$$-mg(y_0 - y_A) = \frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2$$

$$mgy_A + \frac{1}{2} m v_A^2 = mgy_B + \frac{1}{2} m v_B^2$$

$$E_A = E_B$$

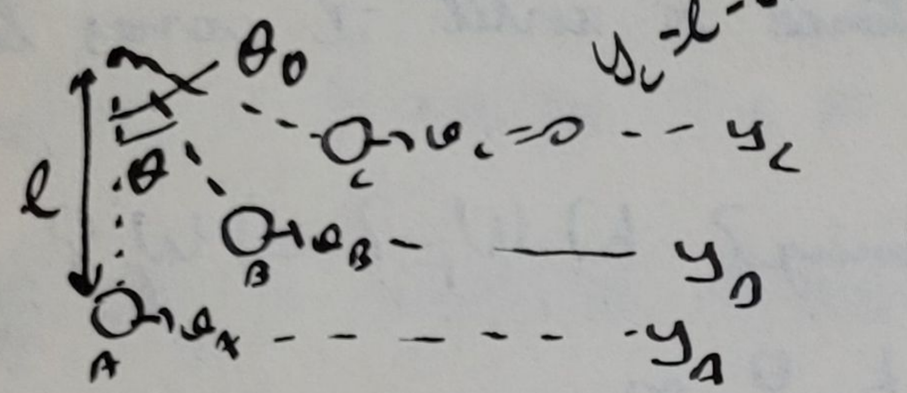
$U_A + KE_A = U_B + KE_B$ = Mechanical energy

$U = mgh$ $\underline{\quad h \quad}$ $2h$ $U = 2mgh$ $\quad -\Delta U = \Delta KE = W_{net}$

ω $\underline{\quad h \quad}$ h $U = 0$

$-mgh$ $\underline{-h}$ $\underline{\quad 0 \quad}$ $U = 0$

- Swinging pendulum



$U_A + KE_A = U_B + KE_B = U_C + KE_C$

$mgy_A + \frac{1}{2}mv_A^2 = mgy_B + \frac{1}{2}mv_B^2$

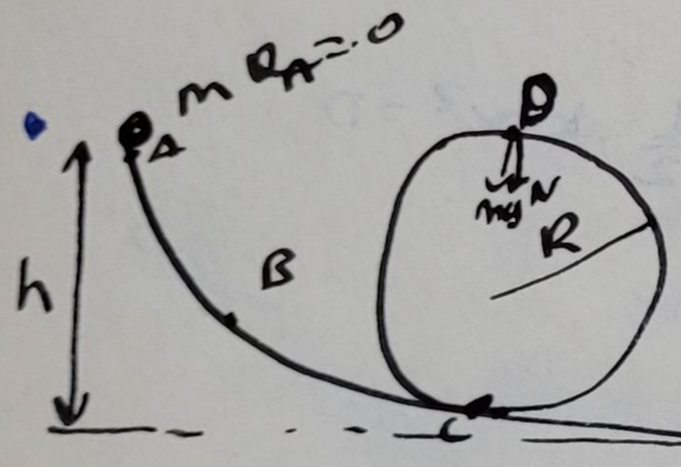
$\hookrightarrow y_B = l - l\cos\theta$

$\frac{1}{2}mv_B^2 = mg(y_C - y_B)$

$\frac{1}{2} \cdot m \cdot v_B^2 = mg(\cancel{l} - l\cos\theta_0 + l\cos\theta - \cancel{l})$

$\frac{1}{2} \cdot \cancel{m} \cdot v_B^2 = \cancel{m}gl(\cos\theta - \cos\theta_0)$

$v_B^2 = 2gl(\cos\theta - \cos\theta_0)$



What is the minimum height h (in terms of R) such that object moves around loop without falling off the loop?

$U_A + KE_A = U_D + KE_D$

$mgh = mg2R + \frac{1}{2}mv_D^2$

$\cancel{m}gh = \cancel{m}g2R + \frac{1}{2} \cdot \cancel{m} \cdot R \cdot \cancel{g}$

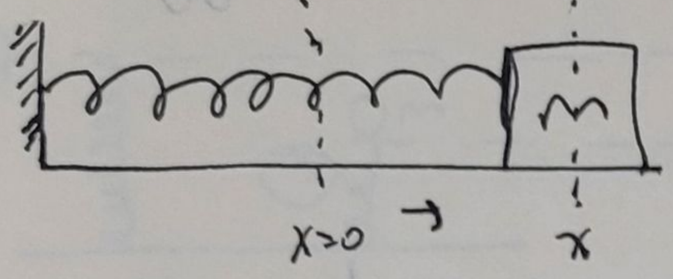
$h = 2R + \frac{1}{2}R = \frac{5}{2} \cdot R$

D// $\vec{N} + mg = m\frac{v_0^2}{R}$

$v_0^2 = R \cdot g$

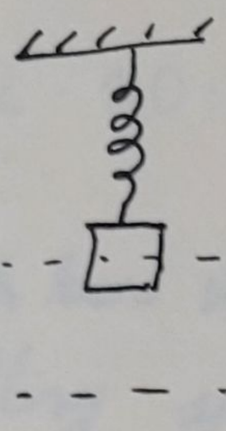
Elastic Potential Energy

28/11/2019



Total mechanical energy

$$E = \frac{1}{2} \cdot k \cdot x^2 + \frac{1}{2} \cdot m \cdot v^2$$

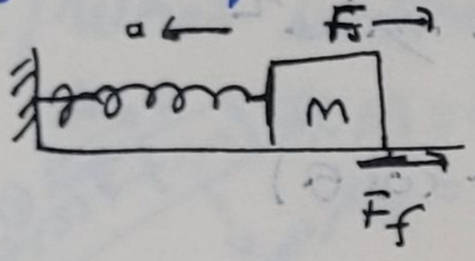


$$W_{net} = W_{grav} + W_{spr}$$

$$= \int_{y_1}^{y_2} \vec{F_G} \cdot \vec{dy} + \int_{y_1}^{y_2} \vec{F_S} \cdot \vec{dy} = KE_2 - KE_1 = -mg(y_2 - y_1) - \frac{1}{2} \cdot k(y_2^2 - y_1^2)$$

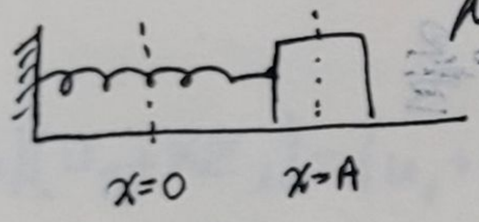
$$\frac{1}{2} m v_2^2 + mgy_2 + \frac{1}{2} \cdot k \cdot y_2^2 = \frac{1}{2} \cdot m v_1^2 + \frac{1}{2} \cdot k \cdot y_1^2 + mgy_1 \qquad E = \frac{1}{2} \cdot mv^2 + \frac{1}{2} kx^2 + mgx$$

- friction force



$$W_{net} = W_S + W_{F_f} = KE_2 - KE_1 = \frac{1}{2} \cdot k(x_1^2 - x_2^2) + W_{F_f}$$

$$\frac{1}{2} \cdot m \cdot v_1^2 - \frac{1}{2} \cdot m \cdot v_2^2 - \frac{1}{2} \cdot k(x_1^2 + x_2^2) = W_{F_f} = -\mu_k \cdot m \cdot g \cdot \Delta x$$



μ_k Calculate velocity of block when it is passing through $x = 0$.

$$W_{F_f} = \frac{1}{2} m(v_2^2 - v_1^2) - \frac{1}{2} \cdot k \cdot (x_2^2 - x_1^2) = \frac{1}{2} \cdot m \cdot v_2^2 - \frac{1}{2} \cdot k \cdot A^2$$

$$\frac{1}{2} \cdot m \cdot v_2^2 = \frac{1}{2} \cdot k \cdot A^2 - \mu_k \cdot mg \cdot A \qquad v_2 = \left(\frac{k}{m} \cdot A^2 - 2\mu_k g \cdot A \right)^{1/2}$$

- force - potential energy relationship

$$W_{net} = \Delta KE = -\Delta u = u_1 - u_2 = \int_1^2 \vec{F} \cdot \vec{dr} \quad \Rightarrow \quad \vec{F} = -\frac{du}{dr} \hat{r} \qquad u(x, y, z)$$

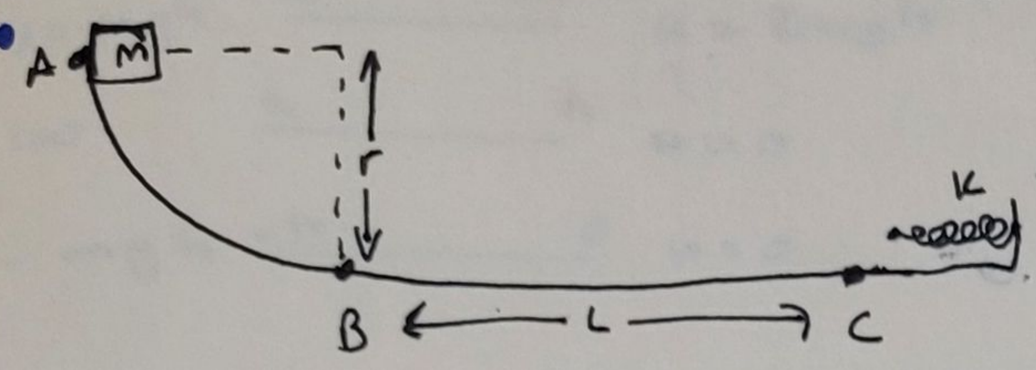
$$\vec{F} = -\frac{d}{dx} u\Big|_{y,z} \hat{i} - \frac{d}{dy} u\Big|_{x,z} \hat{j} - \frac{d}{dz} u\Big|_{y,z} \hat{k}$$

POWER

$$P = \frac{dW}{dt} \qquad \bar{P} = \frac{\Delta W}{\Delta t} : \frac{\text{joule}}{\text{sec}} \triangleq \text{watt} \qquad dW = \vec{F} \cdot d\vec{r}$$

$$P = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

BC → friction μ



a) What is the v_B? b) What is the energy lost?

c) v_C?

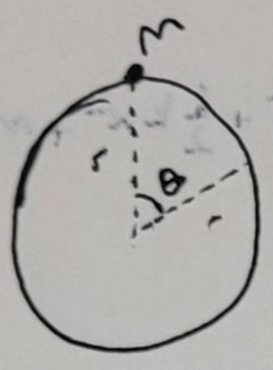
$$mgr = F_f \cdot L + \frac{1}{2} \cdot k \cdot x^2$$

a) $v_B = \sqrt{2gr}$

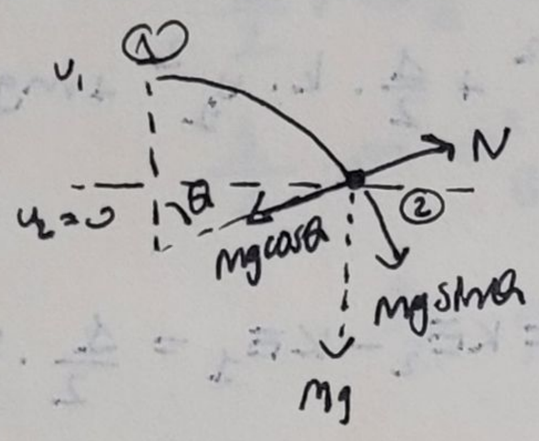
b) $W_{F_f} = -\mu \cdot mg \cdot L$

$$mgr - \mu mgL = \frac{1}{2} m v_C^2 \qquad v_C = \sqrt{2g(r-L)}$$

$$2g(r-L) = v_C^2$$



At what angle will mass leave the sphere?



$$mg\cos\theta - N = \frac{mv^2}{r} \qquad E_1 = E_2$$

$$U_1 = mg(r - r\cos\theta)$$

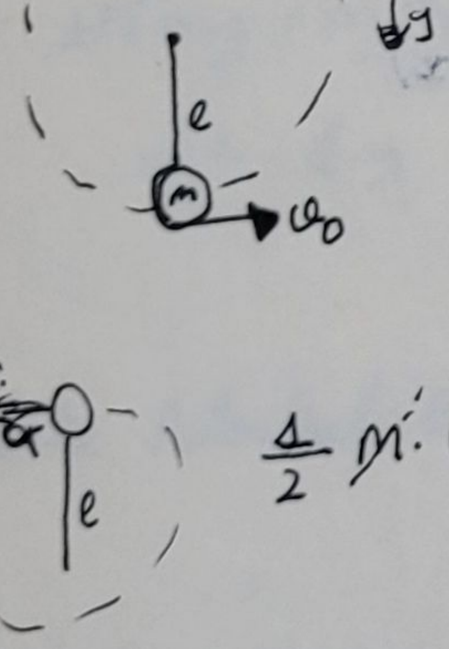
$$mgr(1-\cos\theta) = \frac{1}{2} mv^2$$

$$v^2 = 2gr(1-\cos\theta)$$

$N = 0$ (when mass leave)

$$mg\cos\theta = \frac{mv^2}{r} \qquad v^2 = gr\cos\theta = 2gr(1-\cos\theta)$$

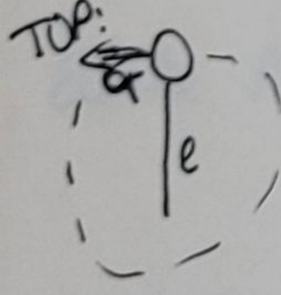
$$\cos\theta = 2 - 2\cos\theta \Rightarrow \cos\theta = \frac{2}{3}$$



a) Find the speed v_0 of the ball at bottom to reach at the top of track barely.

b) Find the tension T is the string at bottom of block in terms of m and g.

TOP:



$$\frac{1}{2} m \cdot v_0^2 = \frac{1}{2} m \cdot v_T^2 + mg2l \qquad v_0^2 = v_T^2 + 4gl = 4gl + gl$$

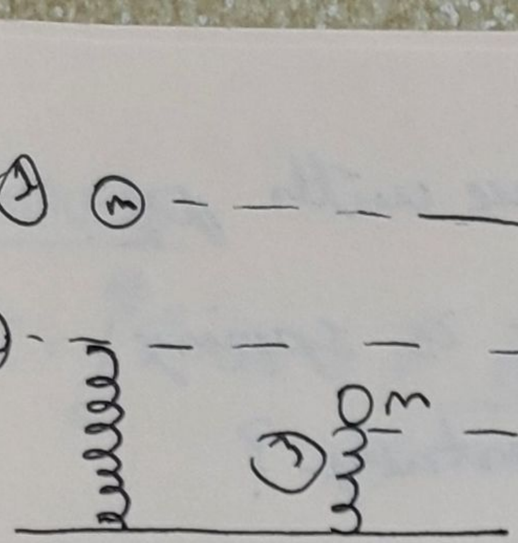
$$mg = \frac{mv_T^2}{l} \qquad gl = v_T^2 \qquad v_0^2 = 4m(1+gl)$$

$$v_0^2 = gl(1+4m)$$

$$\boxed{v_0^2 = 5gl}$$

b) $T_{down} - mg = \frac{mv_0^2}{l}$

$$T_{down} - mg = m\frac{5gl}{l} \Rightarrow \boxed{T_{down} = 6mg}$$



$\frac{1}{2}mg - mg\frac{3}{5} = \frac{1}{2} \cdot k \cdot \frac{1}{100}$

$mg\left(\frac{5}{10} - \frac{1}{10}\right) = \frac{k}{200}$

$\frac{-mg}{10} = \frac{k}{200}$

$m = 2kg \quad y = 10cm \quad h = 50cm \quad k = ?$

$2 \cdot 10 \cdot (0,5) = \frac{1}{2} \cdot k \cdot 10,1^2 + 2 \cdot 10 \cdot (-0,1) \Rightarrow \quad k = 2400 \; N/m$

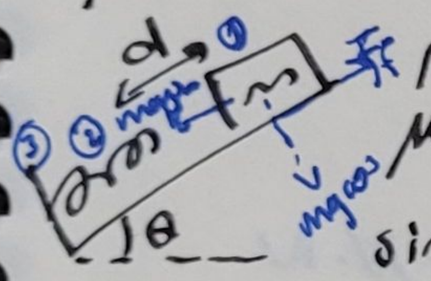
03/12/19

- A box starts from rest and slides down for $d = 1.0m$ before hitting spring.

a) What is the speed of box just before hitting spring?

b) If box compresses spring $\Delta x = 0.2m$ for its equi. pos, $k = ?$

c) What is the maximum value of μ_s between box and surface?



$M = 0.5 \; kg$

$\mu_k = 0.6$

$\sin\theta = 3/5$

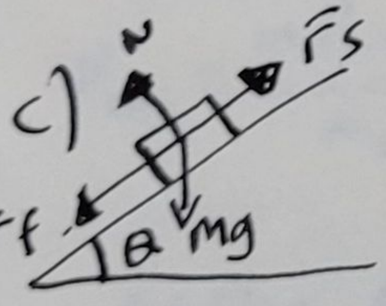
a) $(u_2 + KE_2) - (u_1 + KE_1) = W_{F_f} = -\mu_k \cdot mg \cdot \cos\theta \cdot d$

$\frac{1}{2}mv^2 - mgd\sin\theta = -\mu_k \cdot m \cdot g\cos\theta d$

$\frac{v^2}{2} - 6 = -4 \qquad v = 2 m/s$

b) $(u_3 + KE_3) - (u_1 + KE_1) = -\mu_k \cdot mg \cdot \cos\theta \cdot (d + \Delta x)$

$\frac{1}{2}k\Delta x^2 - mg\Delta x\sin\theta - mgd\sin\theta = -\mu_k mg\cos\theta\,(d + \Delta x) \qquad k = 60 \; N/m$



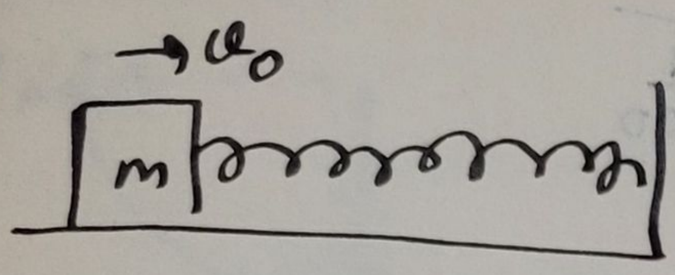
$F_s - F_f - mg\sin\theta = 0$

$k \cdot \Delta x - \mu_s N - mg\sin\theta = 0$

$\mu_s N \geq k\Delta x - mg\sin\theta$

$\mu_s \cdot \frac{5}{10} \cdot 10 \cdot \frac{4}{5} \geq 60 \cdot \frac{2}{10} - \frac{5}{10} \cdot 10 \cdot \frac{3}{5}$

$\mu_s \geq \frac{9}{4}$



Block slides along horizontal surface with μ_k.

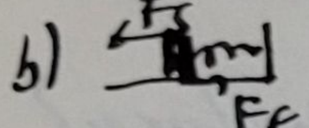
a) if spring constant is k, how far is spring compressed when block stops momentarily?

b) What is the minimum value of μ_s will assure that the spring remains compressed at maximum compressed position?

a) $(u_2 + KE_2) - (u_1 + KE_1) = w_{F_f} = -\mu_k mgx$

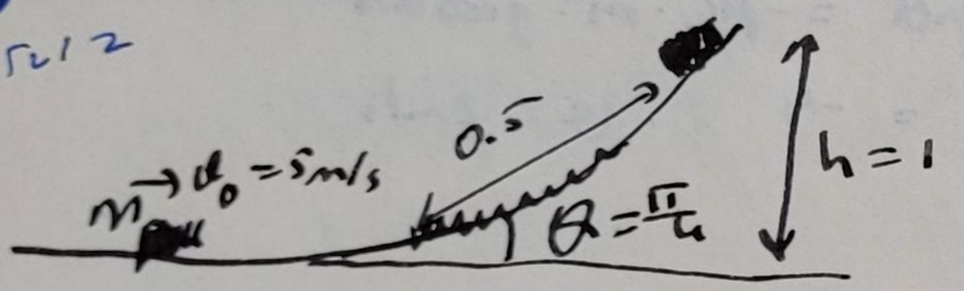
$$\frac{1}{2} \cdot k \cdot x^2 - \frac{1}{2} \cdot m v_0^2 = -\mu_k \cdot mg \cdot x$$

$$kx^2 + 2\mu_k mgx - mv_0^2 = 0$$

b) 

$F_s = F_f \quad k \cdot \Delta x = \mu_s \cdot mg \quad \mu_s = \frac{k \cdot x^2}{mg}$

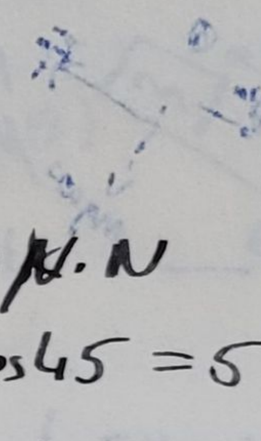
5.12



$EK_0 = \frac{1}{2} \cdot m \cdot v^2 = \frac{1}{2} \cdot m \cdot 25$

$U_2 = mgh = 10m$

$\Delta E = 2{,}5m = W_{F_f} = F_f \cdot \frac{1}{2}$



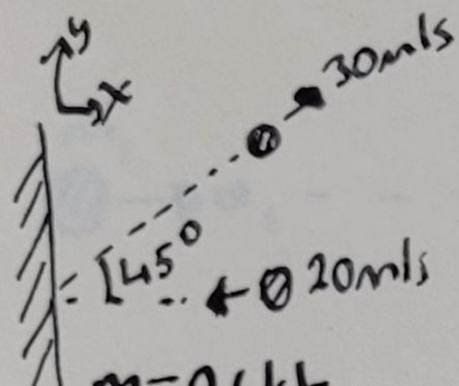
05/12/2019

MOMENTUM- CONSERVATION OF MOMENTUM

10/11/19

IMPULSE

$$I = \int_0^{\Delta t} \vec{F}\,dt = \underbrace{\vec{F}\cdot\Delta t}_{\text{if } F \text{ constant}} = \int_0^{\Delta t} m\frac{d\vec{v}}{dt}\,dt = \int m\,d\vec{v} = \int d(m\vec{v}) = \int d\vec{P} = \Delta P$$



a) What is the impulse of the net force?

b) What is the average net force for a collision time $\Delta t = 0{,}01\,s$?

a) $I_x = \Delta\vec{P} = P_{fx} - P_{ix} = m(v_f\cdot\cos 45 - v_i)$

$= 0{,}4\,(30\cdot\cos 45 - 20) = 16{,}4\ kg\cdot m/s$

$I_y = \Delta\vec{P} = P_{fy} - P_{ix} = m(v_f\cdot\sin 45 - 0)$

$= 0{,}4\,(20\cdot\sin 45) = 8{,}4\ kg\cdot m/s$

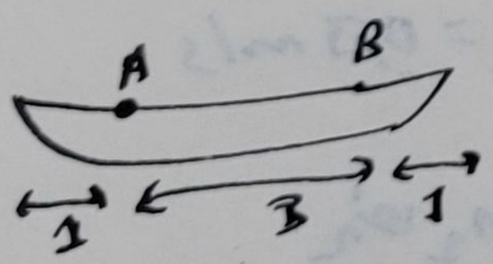
$\vec{I} = 16{,}4\hat{i} + 8{,}4\hat{j}$

b) $F_x = \frac{I_x}{\Delta t} = \frac{16{,}4}{10^{-2}} = 164\,N$ $\quad F_y = \frac{I_y}{\Delta t} = \frac{8{,}4}{10^{-2}} = 84\,N$

Center of mass

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2} \quad \text{if } \vec{F}_{ext} = 0 \text{ center of mass is constant}$$

A 45kg person stands on a 60kg boat of length 5m. Person moves from A to B. How far does the boat move during this process?



$$x_{cm} = \frac{(45\times 1)+(60\times 2.5)}{45+60} = \frac{(45\times 4)+(60\times x_B)}{45+60}$$

$F_{ext} = 0$

$v_{cm} = 0$

$v_{cm} = 0$

$x_{cm} = X\,x_{cm}$

$\frac{45+60\cdot\frac{5}{2}}{} = 45\cdot 4 + 60\cdot x_B$

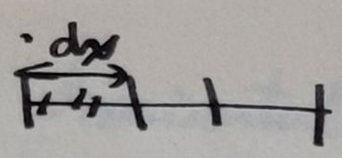
$3+10 = 12+4x_B \Rightarrow 4x_B = 1 \quad x_B = 0.25$

$45+60\cdot\frac{5}{2} = 135 = 45\cdot(4-x) + 60\cdot(2.5-x)$

$x \approx 1.28\,m$ boat moves to the left

$$r_{cm} = \frac{\int r\,dm}{\underbrace{\int dm}_{\text{total mass}}}$$

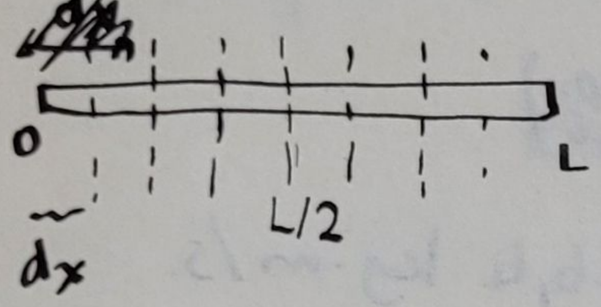
1D: $dm = \frac{M}{L} \cdot dx$



2D: $dm = \frac{M}{A} \cdot dA$

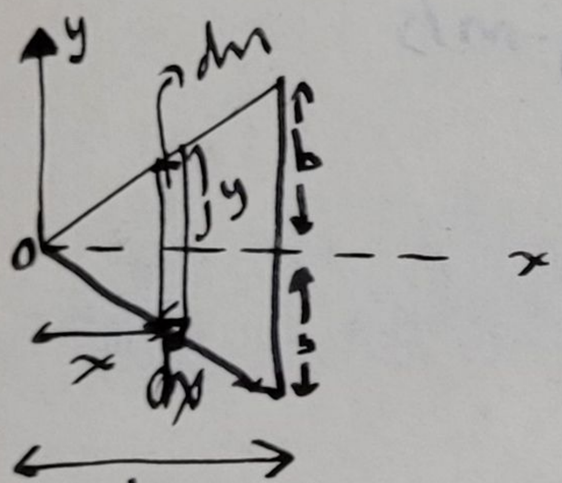
3D: $dm = \frac{M}{V} \cdot dV$ → volume element; V → volume of the object

Center of mass of a this rod M: total mass



$$r_{cm} = \frac{\int_0^L x \cdot \frac{M}{L} \cdot dx}{\int_0^L dm} = \frac{\frac{M}{L}\int_0^L x\,dx}{\int_0^L \frac{M}{L} \cdot dx}$$

$$= \frac{\frac{x^2}{2}\Big|_0^L}{x\Big|_0^L} = \frac{L^2}{2} \cdot \frac{1}{L} = \frac{L}{2}$$



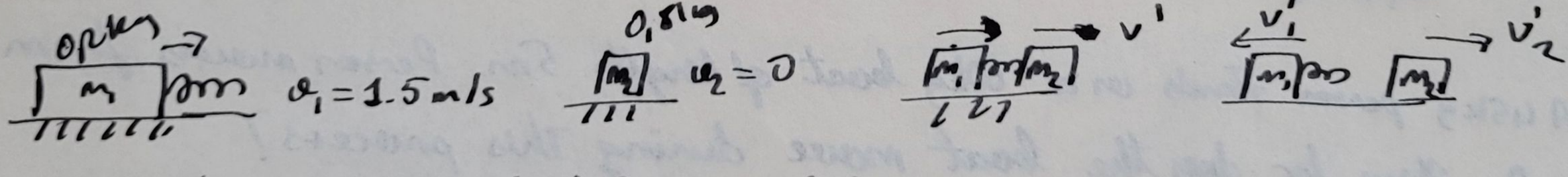
$$x_{cm} = \frac{\int r\,dm}{\int dm} = \frac{\int x \cdot \frac{M}{hb} 2y\,dx}{\int \frac{M}{hb} 2y\,dx}$$

$$\frac{x}{y} = \frac{h}{b} \Rightarrow y = \frac{bx}{h}$$

$$dm = \frac{M}{A} \cdot dA = \frac{M}{h \cdot b} \cdot 2y \cdot dx$$

$$\frac{\int x \cdot \frac{M}{hb} \cdot 2 \cdot \frac{bx}{h} \cdot dx}{\int \frac{M}{hb} \cdot 2 \cdot \frac{bx}{h} \cdot dx} \Rightarrow \frac{\int_0^h x^2\,dx}{\int_0^h x\,dx}$$

$$\frac{\frac{x^3}{3}\Big|_0^h}{\frac{x^2}{2}\Big|_0^h} = \frac{2}{3}h$$



a) What is the center of mass velocity? a) $V_{cm} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{0,2 \cdot 1,5 + 0,8 \cdot 0}{0,2 + 0,8} = 0,3$ m/s

b) Calculate final velocities after collision?

c) Find the max compression if spring $k = 100$ N/m.

b) $P_i = P_f$ $\quad m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$

$$0,2 \cdot 1,5 + 0,8 \cdot 0 = -0,2 v_1' + 0,8 v_2'$$

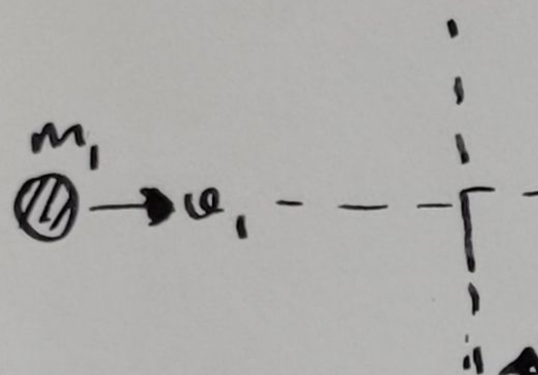
$$0,3 = -0,2 v_1' + 0,8 v_2'$$

$$3 = -2 v_1' + 8 v_2'$$

c) $\frac{1}{2} m_1 v_1^2 = \frac{1}{2}(m_1+m_2)v'^2 + \frac{1}{2} \cdot k \cdot x^2$

$m_1 v_1 = (m_1+m_2)v'$ $\quad x = 0,06m$

$v' = 0,2 m/s$



Masses collide at the origin and mass m_1 stops after collision.

a) Find the velocity of m_2 in terms of given.

b) Find impulses on m_1 and m_2.

c) Find the velocity of center of mass just after collision.

a) $P_i = P_f \quad m_1 v_1 \hat{i} + m_2 v_2 \hat{j} = m_2 \vec{v_2'} \quad v_2 = \frac{m_1}{m_2}\hat{i} + v_2\hat{j}$

b) $I = \Delta P \quad I_{m_1} = m_1 \cdot 0 - m_1 v_1 \hat{j}$

c)